System Integration: Robust Development Stack

1. The Technology "Triad"

- **Frontend:** React (Vite) + TypeScript + Tailwind CSS + Framer Motion.
- **Backend:** Python FastAPI + PostgreSQL + Redis (for API caching).
- **Communication:** RESTful API with Pydantic models for strict type-checking.

2. Embedded Toolkits (Project Dependencies)

The platform must integrate these specific libraries to meet the scientific requirements:

Bio-Logic & Data Handling:

- `Biopython` : For gene sequence analysis and motif searching.
- `Scanpy` : For high-performance single-cell RNA-seq (scRNA-seq) processing.
- `Lifelines` : For Kaplan-Meier and Cox-Proportional Hazards survival models.
- `Pandas` / `NumPy` : For high-speed matrix manipulation.

Statistical Engine:

- `SciPy` & `Statsmodels` : For Mann-Whitney U tests, T-tests, and regression.
- `Rpy2` : To bridge to R-specific packages like `DESeq2` if deeper normalization is required.

Visualization & GUI:

- `ECharts` : For the Expression Atlas and high-performance interactive plots.
- `Cytoscape.js` : For the interactive Protein-Protein Interaction (PPI) networks.
- `NGLView` / `Three.js` : For 3D visualization of protein structures and cleavage sites.

3. API & External Integration

- **GDC API / TCGAbiolinks**: Bulk RNA-seq data.
- **STRING-DB API**: Physical interaction scores.
- **Gemini API**: AI-powered literature synthesis and hypothesis validation.

4. Bilingual Implementation

- **Framework:** `i18next` for React.
- **Storage:** JSON-based locale files (`en.json` , `zh.json`).
- **Rule:** No hardcoded strings in the UI. Everything must pass through the `t()` function.